

Jean Thomas Allen

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EDUCATION

- 2021-present **Ph.D. in Geography**, University of California, Santa Barbara
Thesis: Linking measurements of leaf and canopy water content using imaging spectroscopy time series.
- 2021 **B.S. in Geophysics**, Brown University
Thesis: Phenology of Block Island, RI: Applications of CubeSat Time Series to Pixel Classification of Natural Ecosystems

PUBLICATIONS

2026. **Allen J**, Anderegg LDL, Roberts DA, Trugman AT. Detecting drought stress from the leaf to the landscape: Is canopy structure or leaf desiccation more influential in tree canopy spectral responses to declining water? *New Phytologist*.
2026. **Allen J**, Trugman AT, Anderegg LDL, Boving I, Roberts DA. Variability in leaf water is related to canopy water content over time but not across space in an oak savanna ecosystem. *Remote Sensing of Environment*.
- 2025 Boving I, **Allen J**, Brodrick P, Chadwick KD, Trugman AT, Anderegg LDL (2025). The unstable relationship between drought status and leaf water content complicates the remote sensing of tree drought stress. *Global Change Biology*.
- 2024 Chadwick D, Davis F, Miner K, Pavlick R, Reynolds M, Townsend P, Brodrick P, Ade C, **Allen J**, Anderegg L, Angel Y, Boving I, Byrd K, Campbell P, Carberry L, Cavanaugh K, Cavanaugh K, Easterday K, Eckert R, Gierach M, Gold K, Hestir E, Huemmrich F, Klope M, Kokaly R, Lovegreen P, Luis K, McMahon C, Nidzieko N, Ochoa F, Ongjoco A, Ordway E, Pascolini-Campbell M, Queally N, Roberts D, Saiki C, Schneider F, Shikomanov A, Silva G, Snyder J, Thornton M, Trugman A, Vinod N, Zheng T, Avouris D, Baker B, Baskaran L, Bell T, Berg M, Bernas M, Niklas B, Braghiere R, Breuer Z, Brooks A, Burkard N, Burmistrova J, Cawse-Nicholson K, Chapman J, Charazo-Haraksin J, Cryer J, Cushman K, Dahlin K, Dao P, DiBartolo A, Eastwood M, Elder C, Giodani A, Grant K, Green R, Hanson A, Heberlein B, Helmlinger M, Hook S, Jensen D, Johnson E, Johnson M, Kiper M, Kibler C, King J, Kovach K, Kreisberg A, Lacey D, Lang E, Lee C, Lopez A, Lopez Barreto B, Maguire A, Marsh E, Miller C, Nguyen D, Nickles C, Ocón J, Papen E, Park M, Poulter B, Raiho A, Reim P, Robinson T, Romero Galvan F, Shafron E, Stroschein S, Taylor N, Thompson D, Thompson K, Tye C, Van Beek J, Vanden Heuvel C, Vellanoweth J, Vermeer E, Villanueva-Weeks C, Zumdahl K, Schimel D (2025). Unlocking Ecological Insights from Subseasonal Visible-to-Shortwave Infrared Imaging Spectroscopy: The SHIFT Campaign. *Ecosphere*.

HONORS AND AWARDS

2026	UCSB Graduate Division Dissertation Fellowship (\$18,976)
2026	UCSB Department of Geography Excellence in Research Award
2024	American Geophysical Union Outstanding Student Presentation Award in Hydrology
2023	NSF Graduate Research Fellowship Honorable Mention
2021	Regents Fellowship in Geography – University of California, Santa Barbara (\$41,860)
2017	National Merit Scholarship (\$8000)

GRANTS

2025	UCSB Coastal Fund Grant (\$4,985)
2025	NCALM LiDAR Data Seed Project
2023-2026	NASA Future Investigators in NASA Earth and Space Science and Technology (FINESST) (\$150,000)
2023	Schmidt Family Foundation Research Mentorship Award (\$9,000)
2022-2024	Jack and Laura Dangermond Geography Travel Scholarship (\$1,800)
2020	Undergraduate Teaching and Research Award Summer Research Grant (\$4,000)

PRESENTATIONS

2025	Allen J , Anderegg LDL, Roberts D, Trugman AT, Sensitivity of canopy water content to leaf water potential varies with vegetation structure and plant traits. <i>IEEE International Geoscience and Remote Sensing Symposium (IGARSS)</i> . August 3-8. Oral.
2024	Allen J , P Brodrick, D Thompson. Simultaneous Retrievals of Canopy Water Content and Surface Reflectance in VIS/NIR Spectroscopy. <i>American Geophysical Union Annual Meeting</i> . December 9-13. Poster.
2024	Allen J , Anderegg LDL, Roberts D, Trugman AT. Sensitivity of remotely sensed canopy water content to leaf water potential varies with vegetation structure, hydraulic traits, and plant water status. <i>American Geophysical Union Annual Meeting</i> . December 9-13. Oral.
2024	Allen J , Boving I, Anderegg LDL, Roberts D, Trugman AT. Imaging spectroscopy for monitoring tree water stress: Linking leaf water to canopy water across time and space. <i>Gordon Research Conference on Multiscale Plant Vascular Biology</i> . June 10-14, 2024. Poster.

- 2023 **Allen J**, Trugman AT, Boving I, Anderegg LDL, Roberts DA, Brodrick P, Chadwick D, Jurayj D. Looking forward to SBG: Can Hyperspectral Equivalent Water Thickness Retrievals be used to Monitor Water Stress? *American Geophysical Union Fall Meeting*. December 15. Poster.
- 2023 Jurayj D, Sousa D, **Allen J**, Boving I. Assessing heat stress in oak trees with airborne hyperspectral thermal data. *American Geophysical Union Fall Meeting*. January 22. Poster.
- 2023 Lovegreen P, Chadwick D, Brodrick P, Bohn U, **Allen J**, Boving I, Trugman AT, Anderegg LDL. Investigating the edaphic influence on blue oak water stress with imaging spectroscopy. *American Geophysical Union Fall Meeting*. December 14. Poster.
- 2023 Cantwell T, Sousa D, **Allen J**. Sudden oak death and its forecasted threat to California oaks using maximum entropy modeling. *American Geophysical Union Fall Meeting*. December 12. Poster.
- 2023 Bartleson M, **Allen J**, Sousa D. Differential Drought Impacts on Green Space Across Different Income Areas in Los Angeles, California. *American Geophysical Union Fall Meeting*. December 11. Poster.
- 2023 Boving I, **Allen J**, Lovegreen P, Trugman AT, Anderegg LDL. Remote sensing for ecological insight: linking hyperspectral data products with physiological drought stress. *Ecological Society of America Annual Meeting*. August 10. Oral.
- 2023 Lovegreen P, **Allen J**, Boving I, Brodrick P, Chadwick KD, Kerr K, Trugman AT, Anderegg LDL. Digging deeper: can hyperspectral retrieval of canopy water content detect edaphic influence on blue oak water stress? *Ecological Society of America Annual Meeting*. August 7. Poster.
- 2022 **Allen J**, Trugman AT, Anderegg LDL, Boving I, Roberts DA. Linking remote sensing metrics for water stress with in situ measurements over a 4 month period. *American Geophysical Union Fall Meeting*. December 12. Oral.
- 2022 Schafer N, Sousa D, **Allen J**. Mapping Eucalyptus Mortality Using Hyperspectral Imagery in Ellwood Mesa, Santa Barbara. *American Geophysical Union Fall Meeting*. December 15. Poster.

PROFESSIONAL EXPERIENCE

- 2024 **Graduate Student Intern**, Imaging Spectroscopy Group
Jet Propulsion Laboratory
- 2023 **Research Assistant**, UCSB Planet Labs Data Archive
UCSB Office of Research

- 2022-2023 **Graduate Student Mentor**, Terrestrial Ecology Group
NASA Student Airborne Research Program
- 2019 **Research Assistant**, Icy Moons Freezing Lab
Brown University – Department of Earth, Environmental, and Planetary Sciences

TEACHING EXPERIENCE

Graduate Student Instructor

Remote Sensing of the Environment 3 (Spring 2025)

Teaching Assistant

Introduction to Environmental GIS (Spring 2021)

Glaciers and Climate Change (Fall 2020)

Introduction to Astronomy and Astrophysics (Fall 2018, Spring 2019, Fall 2019, Spring 2020)

SERVICE AND OUTREACH

- 2021-2025 **Python Workshop Helper**, UCSB Data Carpentry
- 2023-2024 **Committee Lead**, Geography Club/UCSB Geography Honors Society Chapter
- 2021-2022 **Classroom Science Visitor**, Foothill Elementary School
- 2021-2022 **Director of Undergraduate Relations**, Geography Equity, Diversity, and Inclusion
Working Group, UCSB
- 2021 **Peer Mentor**, New Scientist Collective
- 2018 **Co-Coordinator**, Brown University Physics Department Undergraduate Group

FIELD EXPERIENCE

- 2022 **Field Assistant**, SBG High-Frequency Timeseries (SHIFT) Field Campaign
Collected leaf area index, transect spectra, and leaf-level spectra
- 2022 **Field Assistant**, Sundowner Winds Experiment (SWEX)
Launched weather balloons carrying radiosondes to measure vertical profiles of
temperature, wind, and relative humidity

RELATED SKILLS

- 2026 **Certificate in Inclusive Teaching**, University of California, Santa Barbara
- 2025 **Certified Carpentries Instructor**, The Carpentries

Programming: Python, R
Languages: English (native), French (conversational)